

Problem Set: Directed Acyclic Graphs

BUSS975: Causal Inference in Financial Research

Four problems. Plan on about three hours. Problem 4 asks you to run a short simulation in R (or Stata); attach your code and output. Nothing else requires software.

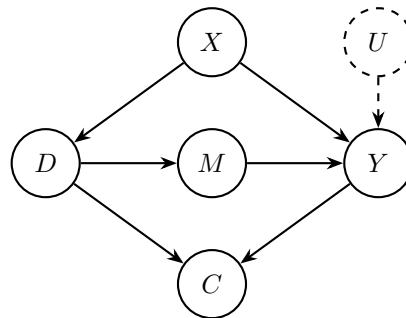
Where a part asks you to explain or evaluate, **answer in at most three sentences**. These are graded on precision, not length.

Further practice — including the material cut from this set — is in the companion file `02_dag_PS_extra.qmd`. It is optional, and good preparation for the comprehensive exam.

Problem 1 (Reading a DAG)

What this is for: everything else in this lecture is an application of path-tracing. Do this one slowly.

Consider the following graph, where U is unobserved.

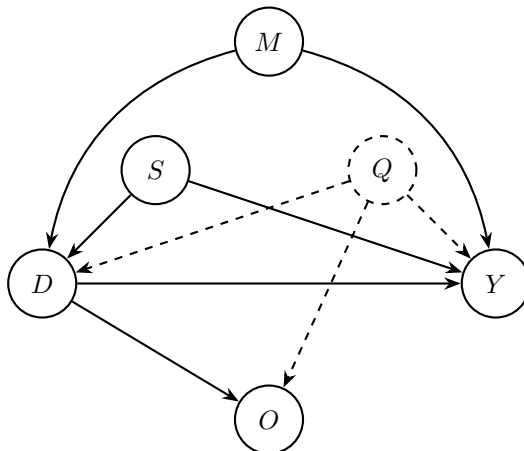


- List **every** path from D to Y and say whether each is directed (causal) or not. For each non-directed path, identify the structure at its middle node — chain, fork, or collider — and state whether the path is open or closed *before* any conditioning.
- Which of these paths are **backdoor** paths? Justify using the definition (both conditions).
- Is C a collider on every path it lies on? Explain why “collider” is a property of a path rather than of a variable.
- Suppose a researcher conditions on $\{X, C\}$. State precisely what each does: which paths open, which close, and whether the causal effect is identified.

Problem 2 (A finance DAG: index inclusion)

What this is for: the previous problem was a graph. This one is a research design — and the answer is that the design fails, which is the point.

You want the effect of **inclusion in a major equity index** (D) on **firm investment** (Y). Let S = firm size, Q = firm quality (**unobserved**), O = institutional ownership, M = market conditions.



- Enumerate every path from D to Y and classify each as directed or backdoor.
- The path $D \rightarrow O \leftarrow Q \rightarrow Y$ connects D and Y . Explain why it is **not** a backdoor path, and why it is nonetheless closed.
- Which set of variables would satisfy the backdoor criterion if everything were observable? Given that Q is *not* observable, is the effect identified by conditioning on observables? State your conclusion and what it implies for the research design.
- A referee insists you control for institutional ownership, O , “since index inclusion changes ownership and ownership affects investment.” Give **two separate** reasons this is wrong.

Problem 3 (Bad controls and selection)

What this is for: the two mistakes that a DAG catches and a regression table never will. Both consist of conditioning on the wrong thing — one deliberately, one by constructing the sample.

- Give the operational rule for identifying a bad control from the DAG, stated in terms of the backdoor criterion. Then explain why a randomly assigned treatment does **not** protect you from bad-control bias.
- For each of the following, identify what the sample-construction step conditions on, and whether that variable is plausibly a collider:
 - A study of mutual fund performance using funds alive at the end of the sample period.
 - A study of the effect of leverage on firm value using CRSP–Compustat merged data.
 - An event study of merger announcements using only **completed** deals.
 - A study of bond pricing using only bonds with a credit rating.

- (c) For each of the following studies, name one plausible bad control and explain the mechanism:
- Effect of a disclosure mandate on cost of capital.
 - Effect of a bank merger on small-business lending.
 - Effect of a governance reform on firm value.
- (d) A paper reports its main specification with and without a questionable control, and the coefficient moves substantially. What should the authors conclude, and what should they *not* conclude?

Problem 4 (Simulation: a bad control with randomized treatment)

What this is for: to see a randomized treatment produce a coefficient of exactly zero when the true effect is 2 — and to be able to predict the number before you run it.

Consider the data-generating process

$$D \sim N(0, 1), \quad A \sim N(0, 1), \quad M = D + A + e_M, \quad Y = \beta D + \gamma A + e_Y$$

with $e_M \sim N(0, \sigma_M^2)$, $e_Y \sim N(0, 1)$, and D, A, e_M, e_Y mutually independent. Take $\beta = 2$, $\gamma = 4$, $\sigma_M^2 = 1$.

- Draw the DAG. Is M a collider, a mediator, or both? Are there any backdoor paths from D to Y ? Before simulating, what do you expect the coefficient on D to be in a regression of Y on D alone, and why?
- Simulate with $n = 200,000$ and report the coefficient on D from three regressions: Y on D ; Y on D and M ; Y on D , M , and A . The second should give approximately **zero** — explain what a researcher who ran only that regression would conclude, and why they would be wrong.
- Derivation.** Show that the coefficient on D in the regression of Y on D and M converges to

$$b_D = \beta - \frac{\gamma}{1 + \sigma_M^2}.$$

Hint: write down $\text{var}(D)$, $\text{cov}(D, M)$, $\text{var}(M)$, $\text{cov}(D, Y)$, $\text{cov}(M, Y)$, then solve the two normal equations. Verify your formula by re-running the simulation at $\sigma_M^2 = 0.25$ and $\sigma_M^2 = 4$.

- Your formula implies the bias **shrinks** as M becomes noisier. State this in words and explain why it is counterintuitive but correct.